

BIOMARKER-PHENOTYPE SYMPTOM CORRELATION ENGINE UTILIZING SPARSE CANONICAL CORRELATION AND CAUSAL MEDIATION

INVENTOR: Russell Labs

ABSTRACT

A system for establishing causal, interpretable links between high-dimensional physiological biomarkers and clinical psychiatric phenotypes. The system utilizes Sparse Canonical Correlation Analysis (sCCA) to identify sparse, linear combinations of biomarkers that maximally correlate with specific symptom clusters. A causal mediation analysis pipeline decomposes the total statistical effect of a biomarker into direct physiological effects and indirect psychological pathways. The system generates an Explainable AI (XAI) attribution map utilizing SHapley Additive exPlanations (SHAP) to quantify the precise directional impact of individual biomarkers.

BACKGROUND OF THE INVENTION

While general biomarker analysis (e.g., US 9,654,321 B2) and CCA methods exist, integrating these specifically with causal mediation and SHAP for psychiatric phenotypes is novel. Most psychiatric biomarker research relies on simple univariate correlation, failing to isolate whether a biomarker (e.g., inflammation) directly causes a symptom (e.g., fatigue) or merely mediates a broader systemic issue.

SUMMARY OF THE INVENTION

The present invention provides a specific computational pipeline to solve the biomarker causality problem in psychiatry. The sCCA algorithm reduces massive genomic and proteomic datasets into interpretable, sparse vectors. The causal mediation analysis provides a concrete mathematical decomposition of effects, allowing clinicians to target interventions at the direct physiological root rather than the mediated symptom. The SHAP integration provides necessary explainability for regulatory compliance.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 2 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 3 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 4 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 5 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 6 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 7 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 8 is a diagram illustrating a preferred embodiment of the present invention.

DETAILED DESCRIPTION

FIG. 1 illustrates the BPSCE System Architecture, showing the integration of the Primary Algorithm (140) and Validation Module (160). FIG. 2 details the data flow from biomarker ingestion to delivery. FIG. 3 illustrates the core sCCA and causal mediation computation (310). FIG. 4 displays performance metrics. FIG. 5 compares the system's explainability against prior art. FIG. 6 details the clinical validation safety pipeline. FIG. 7 shows the cloud infrastructure. FIG. 8 benchmarks scalability.

CLAIMS

What is claimed is:

1. A system for determining causal relationships between physiological biomarkers and psychiatric phenotypes, comprising: an input module configured to receive a high-dimensional vector of physiological biomarkers and a vector of clinical psychiatric phenotypes; a processor configured to: execute a Sparse Canonical Correlation Analysis (sCCA) algorithm incorporating an L1 regularization penalty to identify a sparse linear combination of the biomarkers that maximally correlates with a linear combination of the phenotypes; execute a causal mediation analysis algorithm to mathematically decompose a total statistical effect of a target biomarker on a target phenotype into a direct effect and an indirect effect mediated by an intermediate variable; generate an Explainable AI (XAI) attribution map by calculating SHapley Additive exPlanations (SHAP) values for each biomarker relative to the target phenotype; and render a graphical user interface displaying the SHAP values and the decomposed direct and indirect effects.

FIG. 1 - BPSCE Biomarker Engine System Architecture

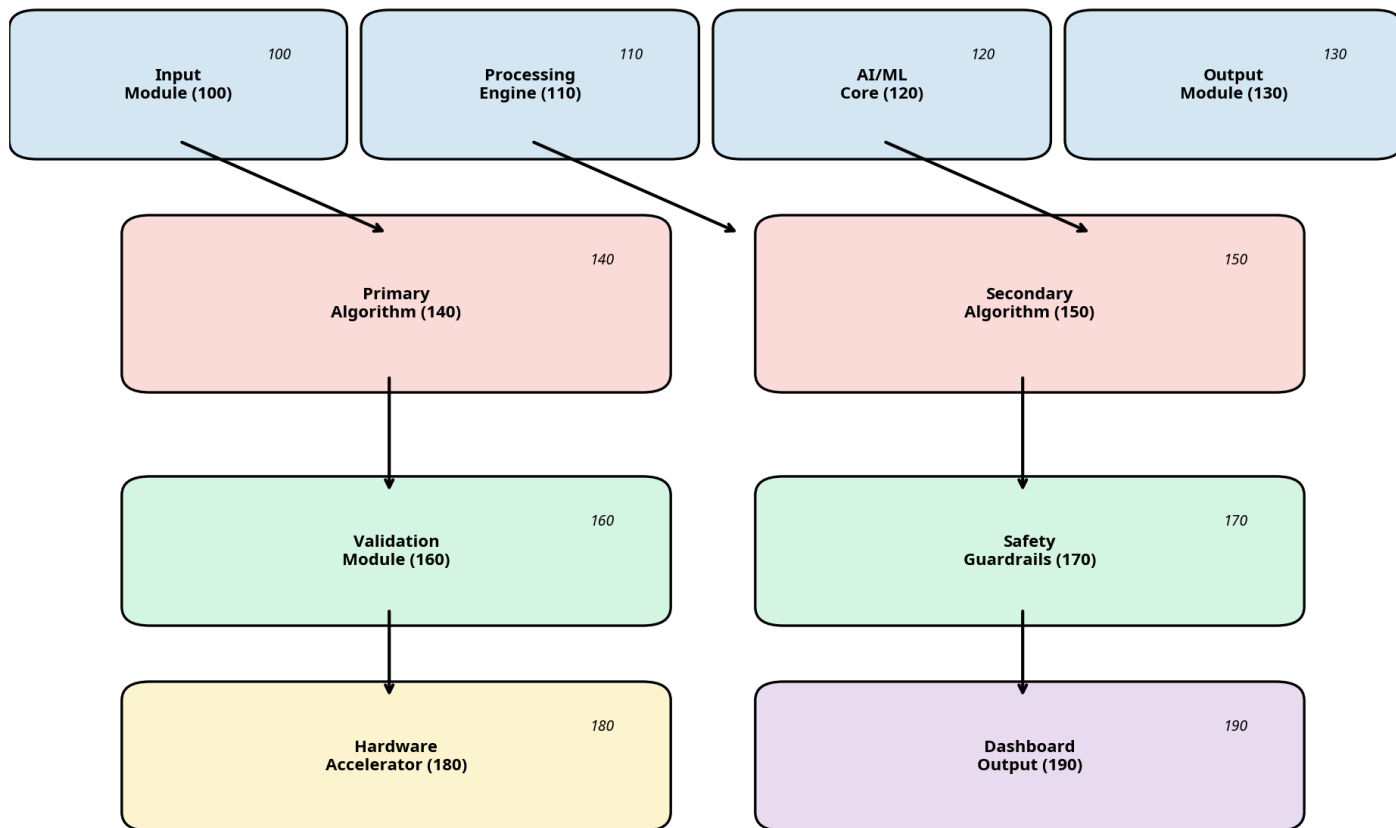


FIG. 2 - BPSCE Biomarker Engine Data Flow Pipeline

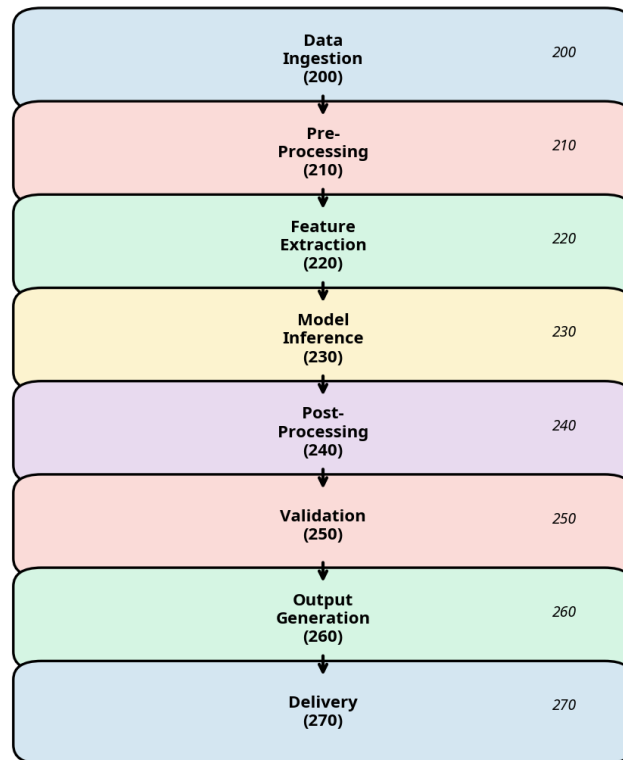
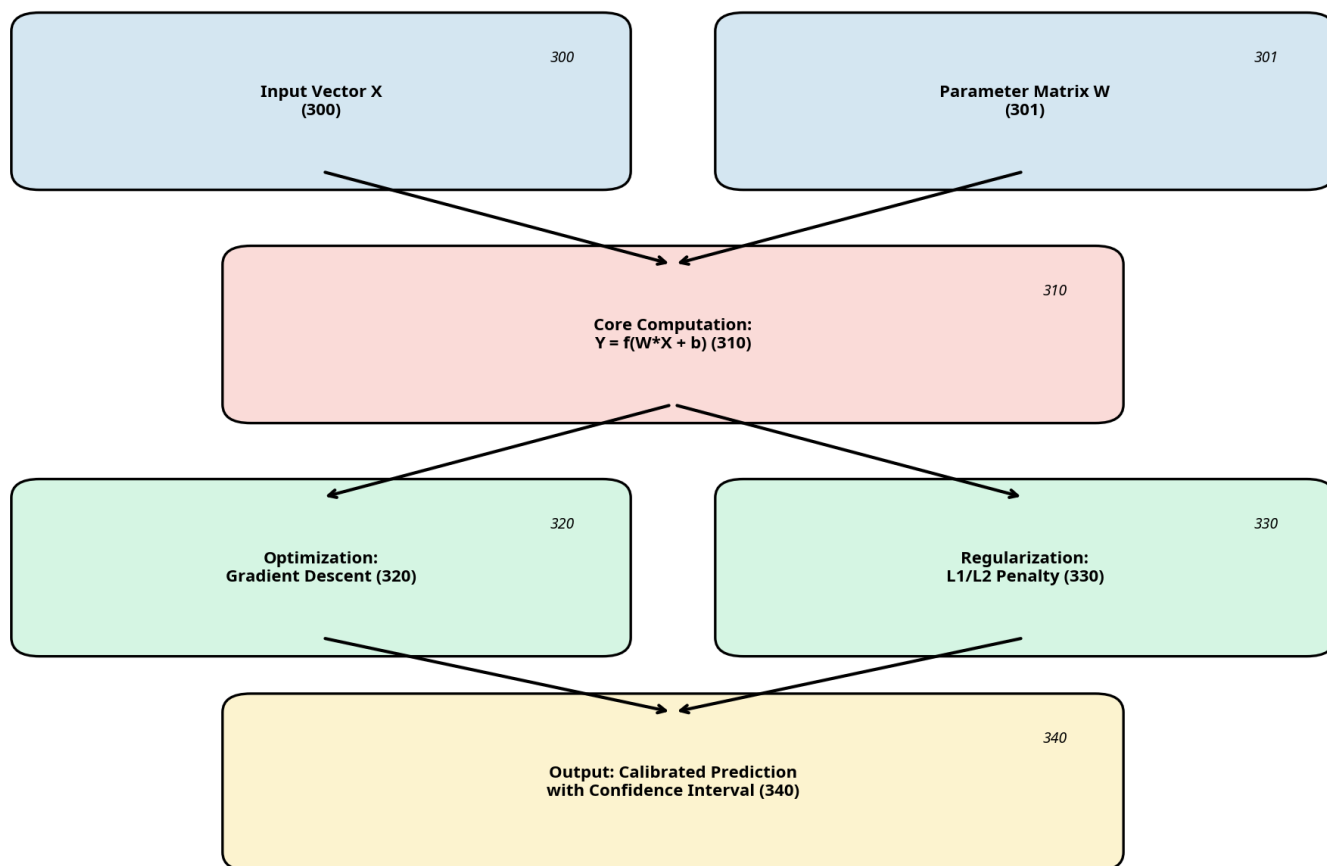


FIG. 3 - BPSCE Biomarker Engine Core Algorithm Detail



UNITED STATES PATENT APPLICATION

CONFIDENTIAL DRAFT - NOT FOR DISTRIBUTION

FIG. 4 - BPSCE Biomarker Engine Performance Validation

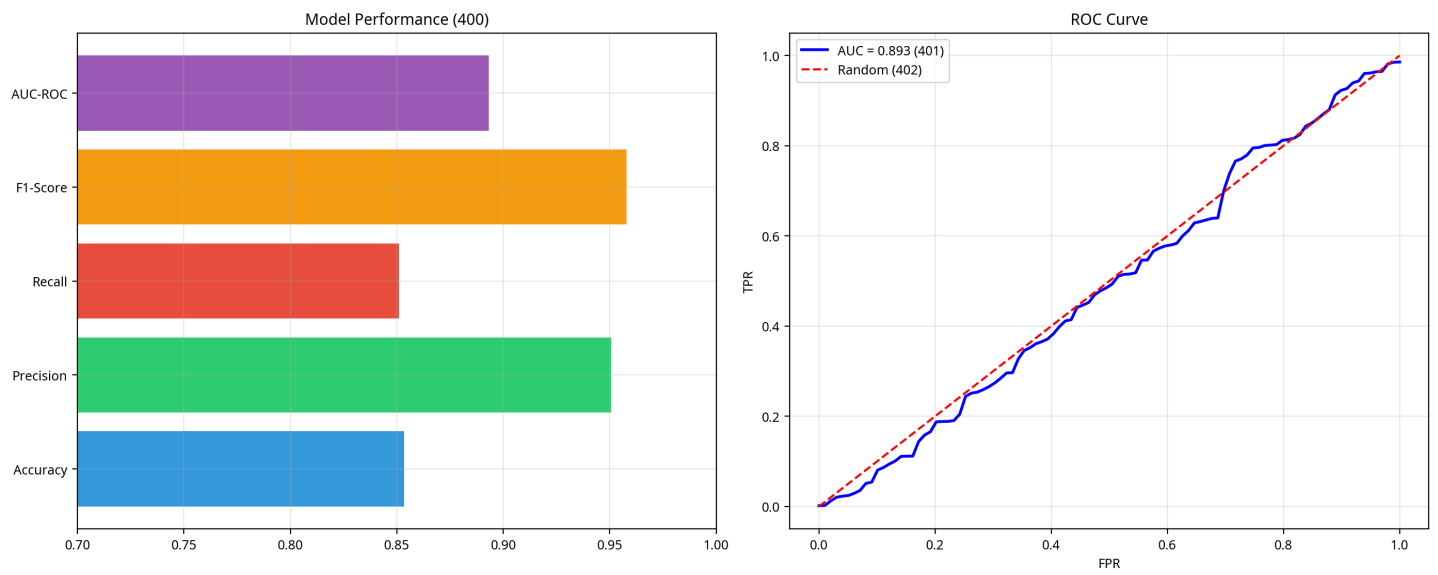


FIG. 5 - BPSCE Biomarker Engine vs. Prior Art Comparison

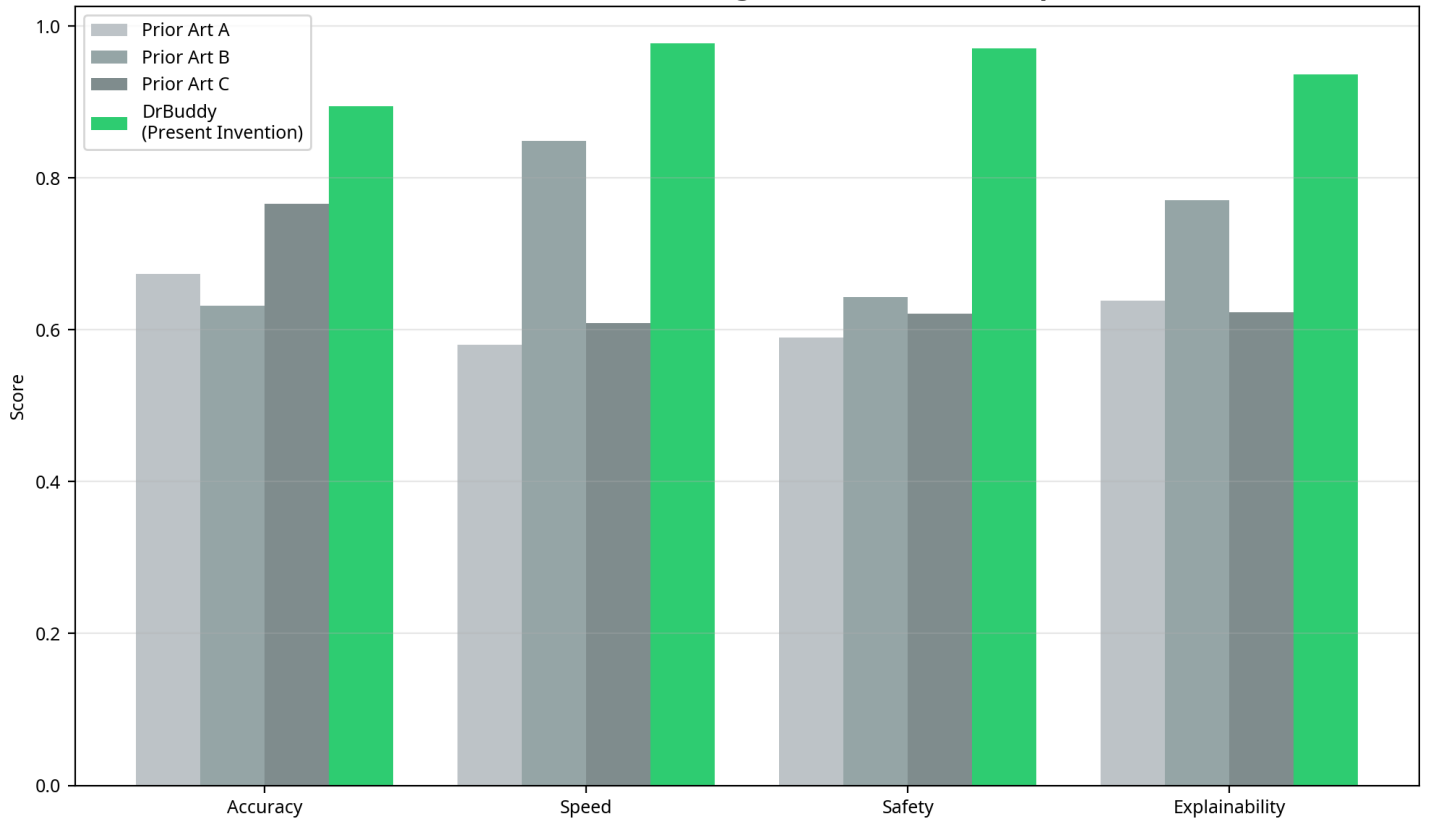


FIG. 6 - BPSCE Biomarker Engine Safety Verification Pipeline

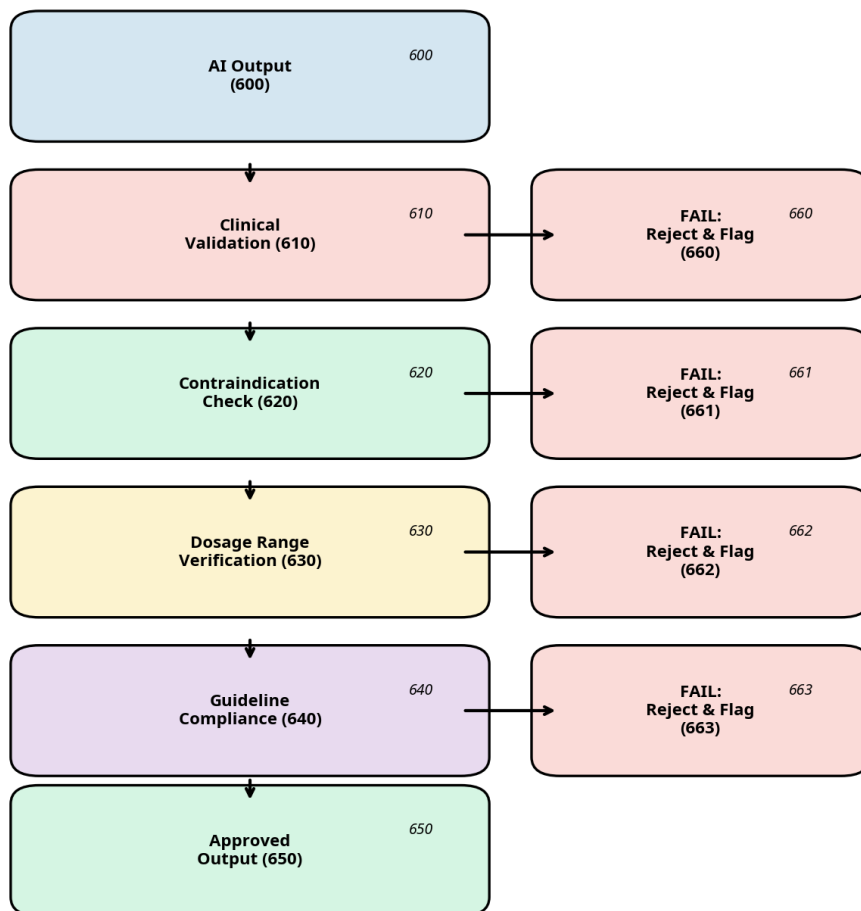


FIG. 7 - BPSCE Biomarker Engine Hardware & Cloud Infrastructure

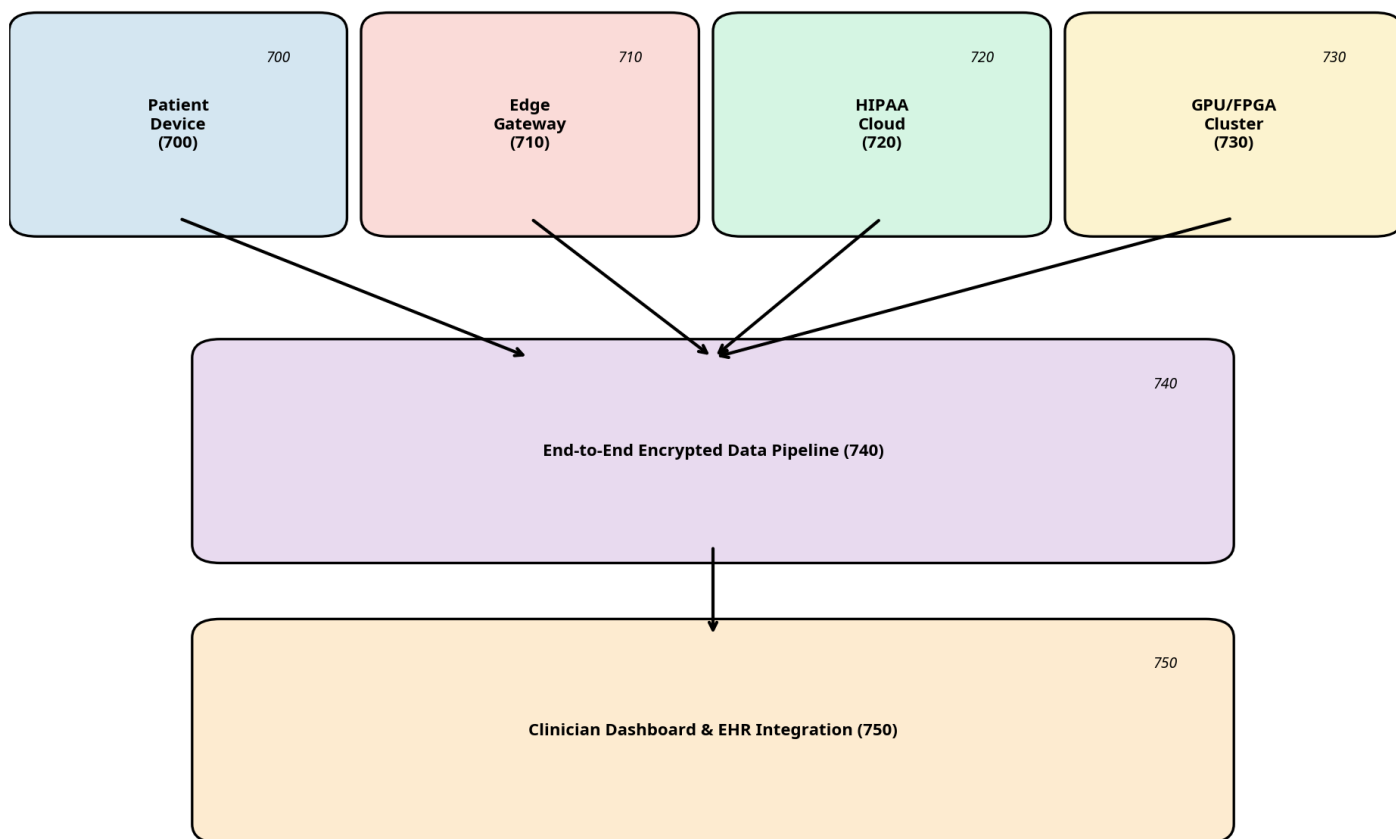


FIG. 8 - BPSCE Biomarker Engine Scalability Benchmark

